The ProbLemma's Channel Season 3 Guide

Season 3 Episode 1: The Index Of Tools

- All the problem-solving approaches studied so far, across seasons 1 and 2, are indexed by the episodes in which they were defined and practiced

Season 3 Episode 2: An Integral Transformation

- Problem S3M1 solved, show that for all θ, x real, with $x > 1$, and $n = 1, 2, 3, 4, 5 \ldots$ it is the case that:

$$\int_0^{\pi} \left(x + \sqrt{x^2 - 1}\cos(\theta)\right)^n \, \mathrm{d}\theta = \int_0^{\pi} \frac{\mathrm{d}\theta}{\left(x - \sqrt{x^2 - 1}\cos(\theta)\right)^{n+1}}$$

- Problems S3M2/3/4 formulated:
  - Problems S3M2/3/4: deduce in at least 3 distinct ways an alternative expression (a product) for the following finite sum:

$$\sum_{k=0}^{n} \cos(a + k\theta)$$

Season 3 Episode 3: One Telescope, Two Telescopes

- Problem S3M2 solved

Season 3 Episode 4: A Complex Approach

- Problem S3M3 solved

Season 3 Episode 5: Euclid Says

- Problems S3M4 solved
- Problem S3P1 formulated:
  - S3P1: find the period of oscillation of a triangular spring oscillator

Season 3 Episode 6: Triangular Spring Oscillator

- Problem S3P1 solved
- Problem S3CS1 formulated:
  - S3CS1: sort 3 integers creatively, without the popular containers and/or industrial strength algorithms

Season 3 Episode 7: 3-Integer Sort, A Discovery

- Problem S3CS1 solved

- Problem S3M5 formulated:
  - S3M5: construct a point, N, inside of a right triangle, ABC, such that the magnitudes of all the angles NAB, NBC and NCA are all equal one another

Season 3 Episode 8: Brocard Points Introduction
- Problem S3M5 solved
- Problem S3M6 formulated:
  - S3M6: invent a way to evaluate finite sums based on a popular approach that is used in combinatorial analysis to prove binomial identities

Season 3 Episode 9: Summation By Double-Counting
- Problem S3M6 solved
- Problem S3P2 formulated:
  - S3P2: find the distance covered by a bouncing material point

Season 3 Episode 10: Infinity In Physics
- Problem S3P2 solved
- Problem S3CS2 formulated:
  - S3CS2: suggest a Turing Machine-friendly algorithm for testing if a given point is located outside or inside of a given triangle

Season 3 Episode 11: A Point-in-triangle Detection Algorithm
- Problem S3CS2 solved
- Problem S3M7 formulated:
  - S3M7: find a way to integrate a binomial differential $x^m(a+bx^n)^p\,dx$

Season 3 Episode 12: Integration Of Binomial Differentials
- Problem S3M7 solved
- Problem S3M8 formulated:
  - S3M8: use the integration of binomial differentials machinery developed in the previous problem in order to evaluate the following infinite sum:

$$\sum_{n=1}^{+\infty} \frac{1}{n(n+1)(n+2)\dots(n+p)}, \quad p \geqslant 2$$

Season 3 Episode 13: Binomial Differentials Slay Infinite Sums Before Breakfast
- Problem S3M8 solved
- Problem S3P3 formulated:
  - S3P3: distinguish a sphere from an identical ball of same radius and mass

Season 3 Episode 14: Sisyphus Versus Spherical Chickens Of Uniform Density

- Problem S3P3 solved
- Problem S3CS3 formulated:
    - S3CS3: construct a single-pass algorithm for sorting **0**s and **1**s

Season 3 Episode 15: Single-pass Janus Sort

- Problem S3CS3 solved
- Problem S3M9 formulated:
    - S3M9: construct an equilateral triangle on 3 parallel straight lines

Season 3 Episode 16: An Equilateral Triangle On 3 Parallel Straight Lines Construction

- Problem S3M9 solved
- Problem S3M10 formulated:
    - S3M10: reconstruct a square given 4 points that belong to one side of that square each

Season 3 Episode 17: 4-Point Square Reconstruction

- Problem S3M10 solved
- Problem S3P4 formulated:
    - S3P4: find the normal and the tangential accelerations as functions of time of a point that moves along a parabola with its acceleration vector staying parallel to the y-axis at all times

Season 3 Episode 18: A Parabolic Motion With Acceleration Staying Parallel To The y-axis

- Problem S3P4 solved
- Problem S3CS4 formulated:
    - S3CS4: creatively swap 2 sub-arrays

Season 3 Episode 19: A Creative Swap Of Two Sub-Arrays

- Problem S3CS4 solved
- Problem S3M11 formulated:
    - S3M11: evaluate a finite trigonometric sum

$$\sum_{k=1}^{n} \frac{1}{\sin\left(2^k x\right)}$$

Season 3 Episode 20: A Finite Trigonometric Sum Evaluation

- Problem S3M11 solved
- Problem S3M12 formulated:
    - S3M12: evaluate a finite binomial sum using Physics:

$$\sum_{k=0}^{n} k \cdot \binom{n}{k}$$

Season 3 Episode 21: Physics Evaluates A Finite Binomial Sum

- Problem S3M12 solved
- Problem S3P5 formulated:
  - S3P5: recover the shape of the trajectory of a point that moves in a plane in such a way that its both tangential and normal accelerations remain constant at all times

Season 3 Episode 22: Planar Motion With Constant Tangential And Normal Accelerations

- Problem S3P5 solved
- Problem S3CS5 formulated:
  - S3CS5: explain how the Shunting Yard Algorithm works and create a Slow-Motion Player that lets a user experiment with and witness the mechanics of that algorithm

Season 3 Episode 23: The Mechanics Of The Shunting Yard Algorithm

- Problem S3CS5 solved
- Problem S3M13 formulated:
  - S3M13: evaluate the following integrals in at least 3 different creative ways:

$$C_{a,b}(x) = \int e^{ax} \cos\left(bx\right) dx; \qquad S_{a,b}(x) = \int e^{ax} \sin\left(bx\right) dx$$

Season 3 Episode 24: 3 Creative Evaluations Of Ubiquitous Integrals

- Problem S3M13 solved
- Problem S3M14 formulated: a divisibility by 7 test
  - S3M14: design a divisibility-by-7 algorithm using the Reduce-And-Conquer approach

Season 3 Episode 25: A Divisibility-By-7 Test Design Via Reduce-And-Conquer

- Problem S3M14 solved
- Problem S3P6 formulated: a kinematic descend of a rotating line segment
  - S3M14: recover the law of motion, the trajectory, the velocity, the acceleration and the curvature of the trajectory traced by one edge of a uniformly descending line segment that also rotates in the horizontal plane with a constant angular velocity

Season 3 Episode 26: A Kinematic Descend Of A Rotating Line Segment

- Problem S3P6 solved
- Problem S3CS6 formulated: describe the mechanics and then design and implement the Narayana's Next-Linear-Permutation algorithm

Season 3 Episode 27: Narayana's Next Linear Permutation Algorithm

- Problem S3CS6 solved
- Problem S3M15 formulated. Evaluate the following finite trigonometric sum:

$$\sum_{m=1}^{n} \arctan\left(\frac{2m}{m^4 + m^2 + 2}\right)$$

Season 3 Episode 28: In Like A Lion Out Like A Lamb

- Problem S3M15 solved
- Problem S3M16 formulated. Evaluate the following limit:

$$\lim_{n \to +\infty} n^{\frac{1}{n}}$$

Season 3 Episode 29: The Art Of Manipulation Of Inequalities

- Problem S3M16 solved
- Problem S3P7 formulated: deduce an equation that describes a spherical motion of a point under a constant angle with the meridians

Season 3 Episode 30: A Deduction Of An Equation Of A Loxodrome

- Problem S3P7 solved
- Problem S3CS7 formulated: construct an O(n) time and O(1) space complexity algorithm for detecting if a given integer can be represented as a sum of two integers from a given array sorted in some order, ascending or descending

Season 3 Episode 31: An Integer As A Sum Of Two Elements Of A Sorted Array

- Problem S3CS7 solved
- Problem S3M17 formulated: evaluate the following indefinite integral integral

$$\int \sqrt{x^2 - a^2}\, dx$$

in at least two creative ways

Season 3 Episode 32: Hyperbolically Speaking

- Problem S3M17 solved
- Problem S3M18 formulated: evaluate the following definite integrals

$$\int\limits_0^{\frac{\pi}{2}} \sin^m x\, dx \quad \text{and} \quad \int\limits_0^{\frac{\pi}{2}} \cos^m x\, dx$$

for $m = 1, 2, 3, 4, 5$ and so

Season 3 Episode 33: Recurrence Relations Evaluate Integrals

- Problem S3M18 solved
- Problem S3M19 formulated: show that hyperbolic functions can be defined by a direct analogy with circular functions defined via origin-centered unit circle

Season 3 Episode 34: Hyperbolic Functions Defined By A Direct Analogy With Circular Functions

- Problem S3M19 solved
- Problem S3P8 formulated: recover the law of motion of a point that is moving along a straight line that sweeps the surface of a right circular cone

Season 3 Episode 35: Twice Telescopic Rotating Antenna, The Kinematics Of

- Problem S3P8 solved
- Problem S3CS8 formulated: construct an algorithm for detecting a cycle in a linked list

Season 3 Episode 36: A Vicious Circle Or Lords Of The Ring

- Problem S3CS8 solved
- Problem S3M20 formulated: show that for all real $x \neq \pm k\pi$, where $k = 0, 1, 2, 3$ and so on, it is the case that:

$$\sin(x) = x \cdot \prod_{n=1}^{+\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Season 3 Episode 37: Sine As A Product, With Applications

- Problem S3M20 solved
- Problem S3M21 formulated: evaluate the Euler-Poisson integral

$$\int_0^{+\infty} e^{-x^2}\, dx$$

using the Squeeze Theorem.

Season 3 Episode 38: What Mathematicians Do All Day Or Euler-Poisson Integral Evaluation via the Squeeze Theorem

- Problem S3M21 solved
- Problem S3P9 formulated: show

Season 3 Episode 39: Lorem Ipsum

- Problem S3P9 solved
- Problem S3CS9 formulated: show

Season 3 Episode 40: Lorem Ipsum

- Problem S3CS9 solved
- Problem S3M22 formulated: show

Season 3 Episode 41: Lorem Ipsum

- Problem S3M22 solved

- Problem S3M23 formulated: show

Season 3 Episode 42: Lorem Ipsum
- Problem S3M23 solved
- Problem S3P10 formulated: show

Season 3 Episode 43: Lorem Ipsum
- Problem S3P10 solved
- Problem S3CS10 formulated: show

Season 3 Episode 44: Lorem Ipsum
- Problem S3CS10 solved
- Problem S3M24 formulated: show

Season 3 Episode 45: Lorem Ipsum
- Problem S3M24 solved
- Problem S3M25 formulated: show

Season 3 Episode 46: Lorem Ipsum
- Problem S3M25 solved
- Problem S3P11 formulated: show

Season 3 Episode 47: Lorem Ipsum
- Problem S3P11 solved
- Problem S3CS11 formulated: show

Season 3 Episode 48: Lorem Ipsum
- Problem S3CS11 solved
- Problem S3M26 formulated: show

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S3CS1	Season 3 Episode 6	Season 3 Episode 7
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